# When can you trust a topological visualization?

A task-based evaluation of the Scikit-TDA ecosystem (ripser, persim) and KeplerMapper

Shourya Sarkar · Scientific Visualization — Evaluation of Visualization Systems · 28 October 2026

# A topological picture is not the topology

| The data's true topology | Persistent homology | Mapper graph | The drawing |
| --- | --- | --- | --- |
| What the shape really is. Usually unknown; known here only for synthetic data. | Evidence under one metric and filtration. A dominant loop means $r_1 \geq 3$, a convention, not a theorem. | Depends on lens, cover, clusterer and its scale. A graph cycle is not a loop in the data. | Layout and encodings can hide components or suggest proximity without changing the graph. |

- A fifth factor this study cannot measure: how a person reads the picture. No participant study was run.
- Every claim is tied to one of these four objects.

# What was evaluated, and how

- **System:** ripser 0.6.15, persim 0.4.1, kmapper 2.1.0; scikit-learn, networkx and matplotlib counted as external
- **Core experiments:** E1 capability audit (13 tasks) · E2 Mapper parameter grid (24 cells + no-loop control) · E3 single-observer inspection
- **Extension studies:** weather case study · offline Mapper explorer · resampling robustness
- **Controls:** clusterer, null series, independent redraws, and two quantitative caveat checks
- **Data with known answers:** annulus (1 loop), figure-eight (2), blob (none); plus NOAA station data 2019–2024

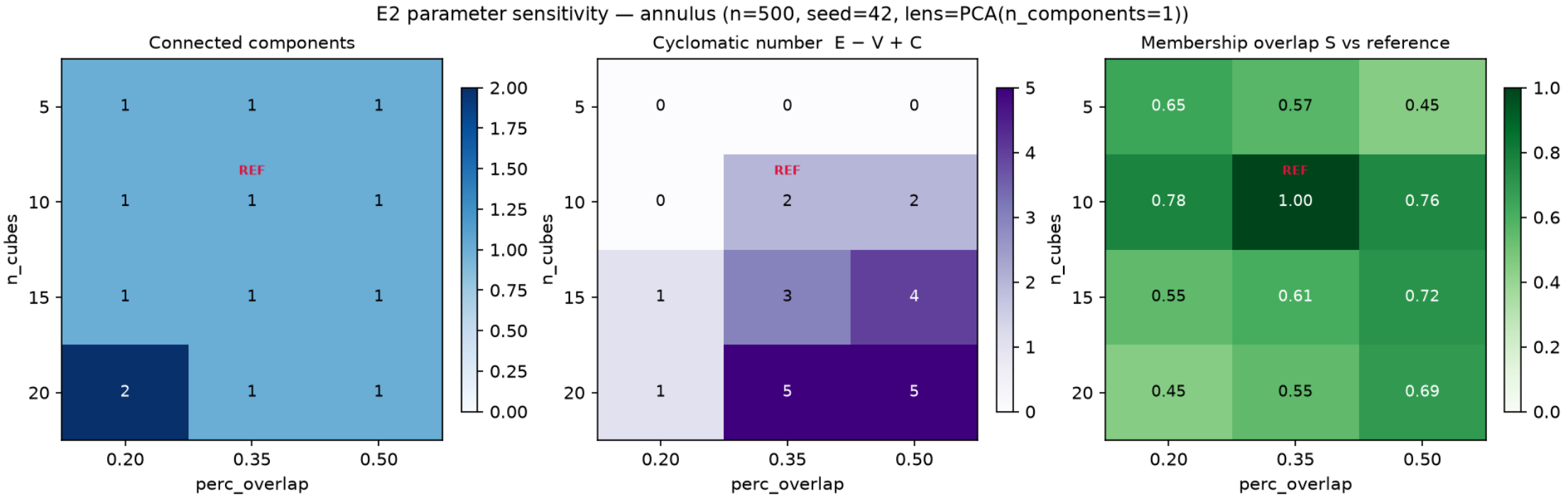
# E1: the ecosystem computes; you assemble the rest

| Native | Your code | Other packages | Not offline |
| --- | --- | --- | --- |
| install · persistent homology · diagram plot · bottleneck/Wasserstein · matching plot · static Mapper drawing | data generators · H1 barcode · reproduction harness. 130 counted glue lines in total | scikit-learn: lens and clustering · networkx: component and cycle counts · PCA baseline | KeplerMapper HTML needs D3, FileSaver and a web font from CDNs: 0 nodes drawn without a network |

- Static Mapper layout can overlap disconnected components (bounding-box overlap 0.23 on three blobs)
- ripser 0.6.15 has no plotting; persim draws diagrams, not barcodes

# E2: cover parameters change the Mapper story

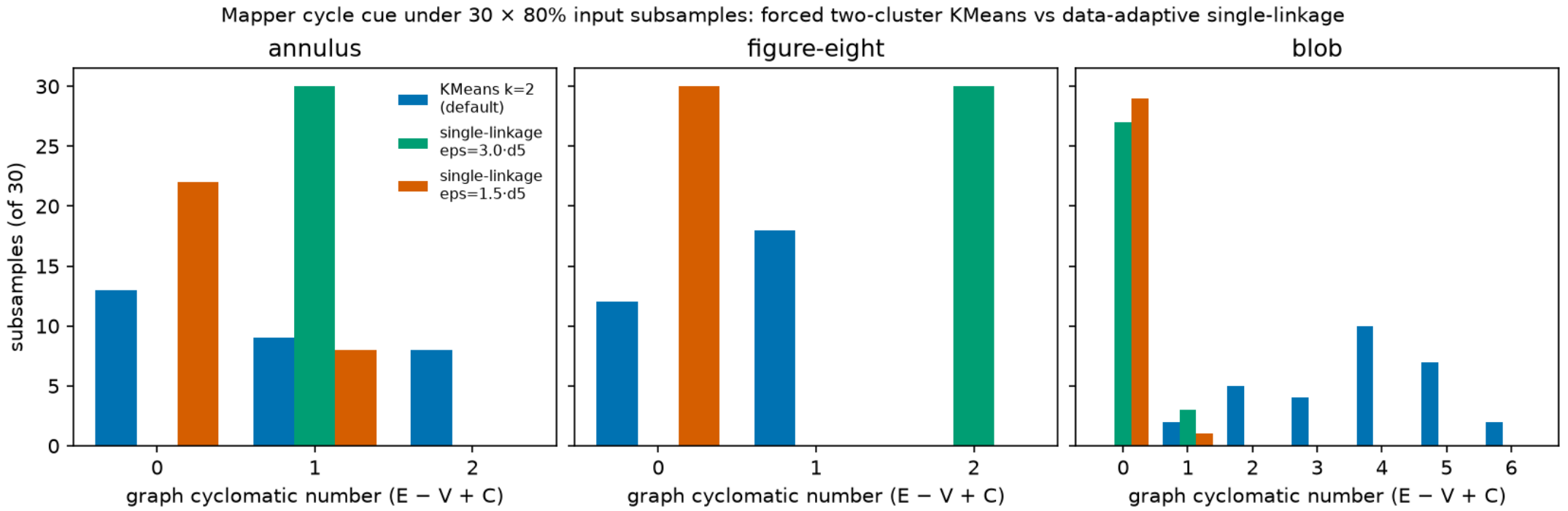
- Annulus: tree at 5 intervals, connected with 2 cycles at (10, 0.35), fragments at (20, 0.20); figure-eight: 2 components at the reference cell
- Node count = 2 × n_cubes in all 24 cells — because the clusterer is forced to make two clusters per interval
- No-loop blob: graph cycles, but no dominant H1 class




Annulus, 12 cells: components, cycles and membership overlap S by cover resolution and overlap

# The clustering rule and scale decide whether the cycle cue survives

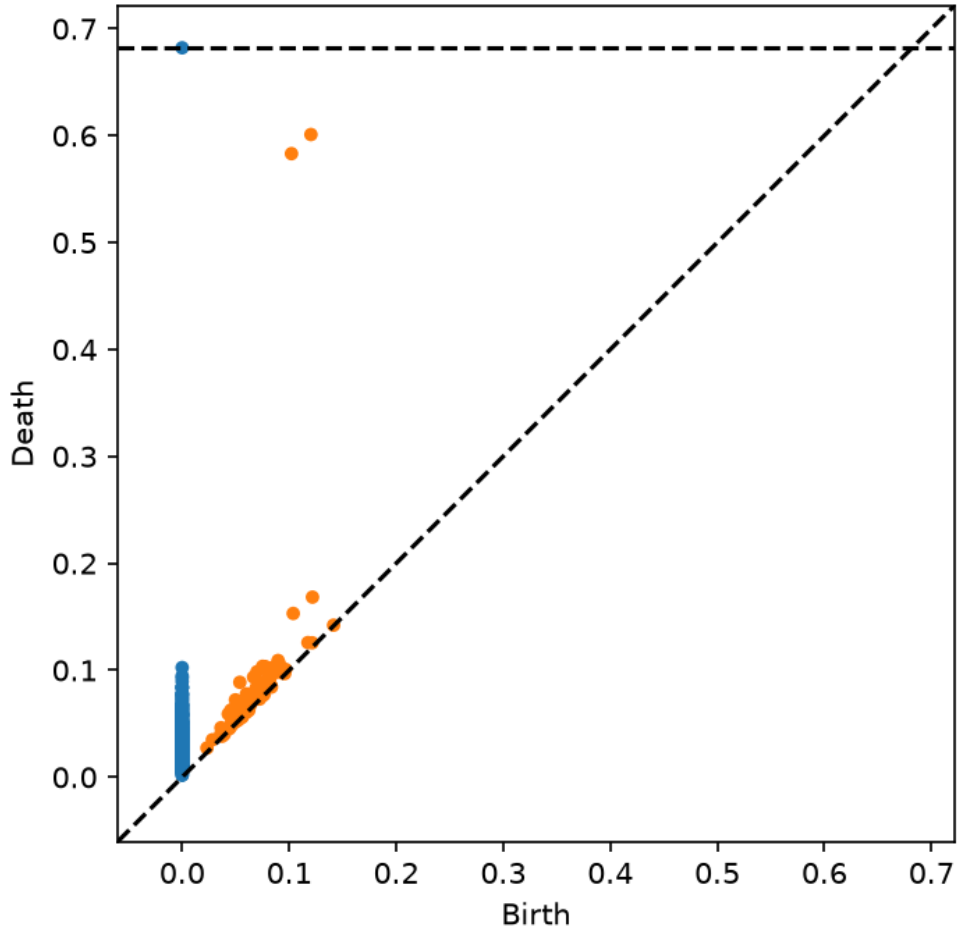
- Annulus, KMeans k=2: no cycle in 13/30 subsamples and 20/30 independent redraws. Single-linkage at 3× the median 5-NN distance: exactly one cycle in 30/30 and 30/30. At 1.5×: absent in 22/30, graph fragments.
- The two scales were fixed after the KMeans result was known and only these two were tried: an exploratory existence demonstration, not a recommended or independently confirmed setting.




Cyclomatic number of the Mapper graph over 30 subsamples (80%) per shape: forced two-cluster KMeans vs single-linkage at two scales

# E3: what the artifacts communicate — one observer, not a study

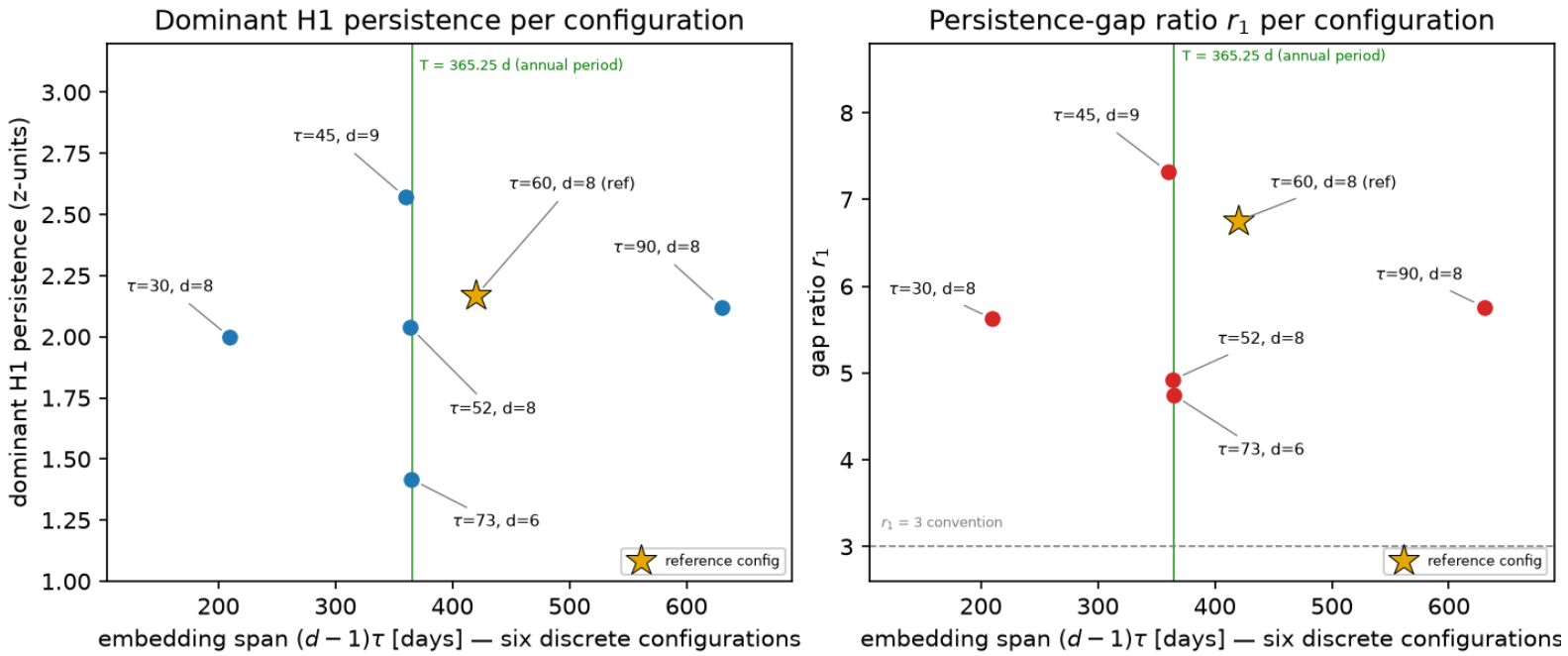
- No participant study was conducted; no participant data exist.
- 13 stimuli inspected by the author with an AI vision model as viewing aid: 11 clearly apparent, 2 ambiguous
- Ambiguities concern encodings: unlabeled H0/H1 colours and the essential class; noise-scale Mapper cycles that look like a ring
- Evidence about artifacts, not about viewers




Stimulus S08 as inspected: no legend; the essential H0 point (top) sits on the dashed line

# Weather: a strong H1 cue in a temporal embedding

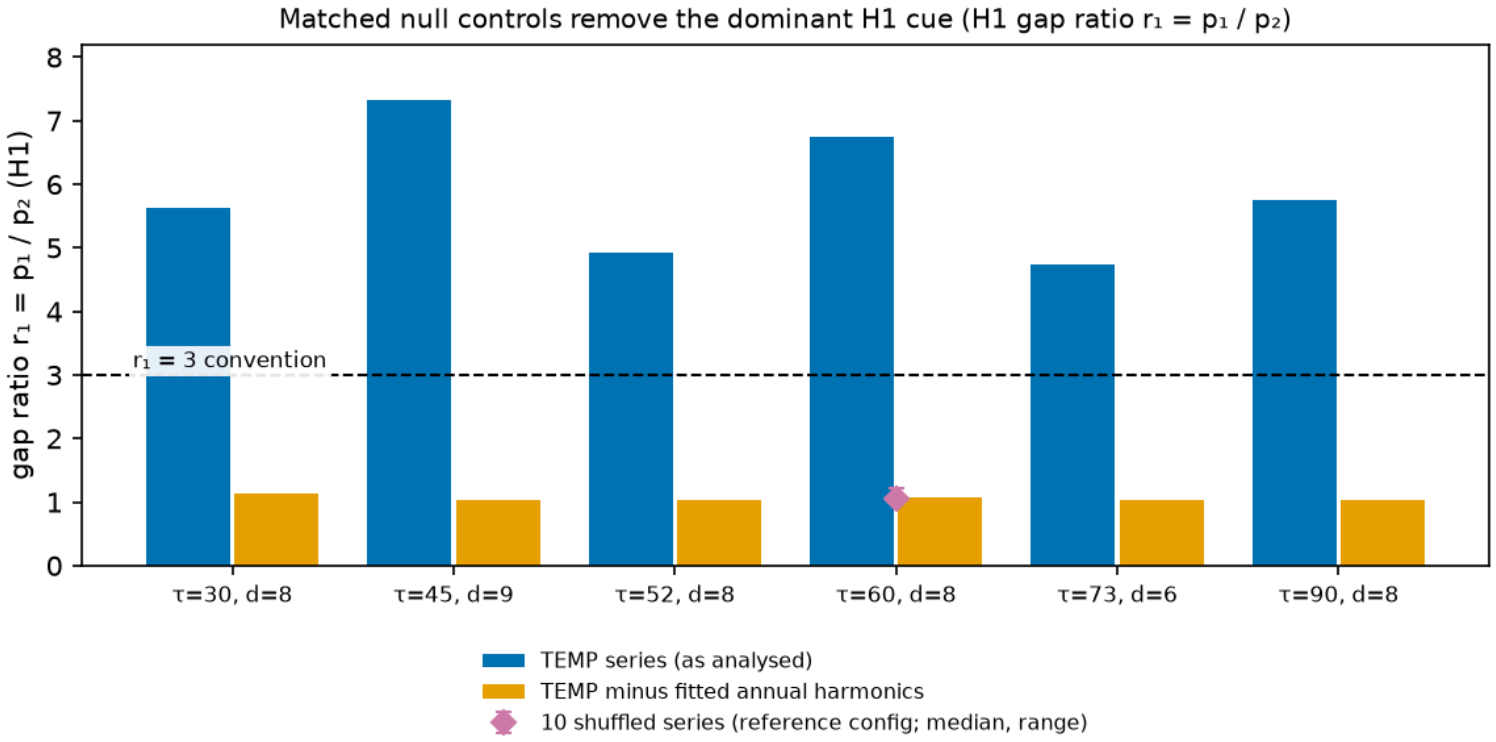
- NOAA GSOD station 72505394728; raw files verified by hash. Embedding of daily TEMP: one dominant H1 class in all six configurations, $r_1$ = 4.75–7.32
- The six share one dataset and were designed around the annual period — consistency, not replication
- Static 3-D state cloud: $r_1$ = 1.19 — but that is an illustration, not a null control




Six sliding-window configurations of daily TEMP, Central Park, 2019–2024 (2192 days)

# Matched null controls remove the dominant H1 cue

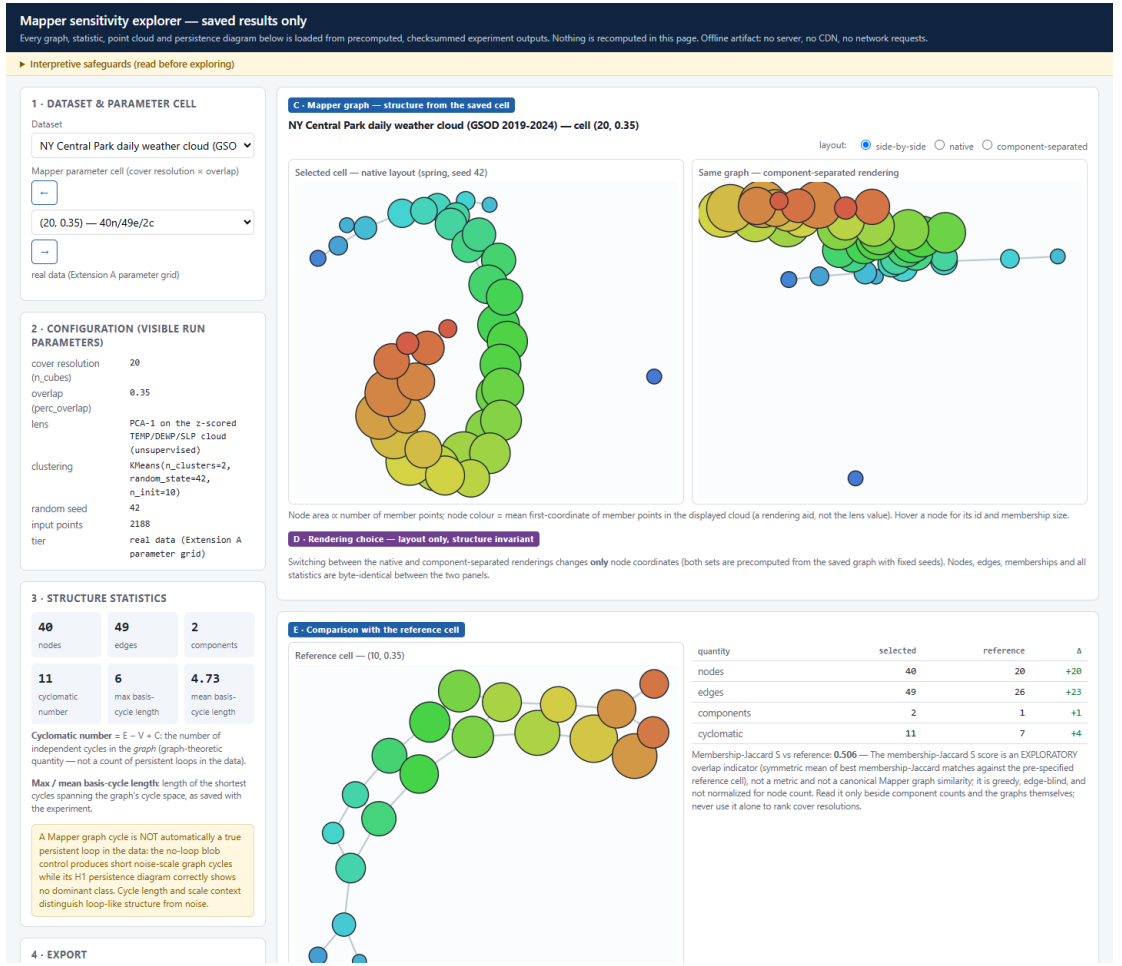
- Same embedding and configurations. Two annual harmonics explain 82.2% of the variance. Without them $r_1$ = 1.02–1.14; ten shuffles: 1.01–1.22 — none meets $r_1 \geq 3$.
- H1 classes remain in the null embeddings; the dominance criterion is not met. This does not show the annual cycle is the only source, or that no loop exists.




H1 gap ratio $r_1$ for the TEMP series, the anomaly series (two annual harmonics removed), and 10 shuffled series; dashed line = $r_1$ = 3

# Offline explorer: inspect parameter dependence without a server

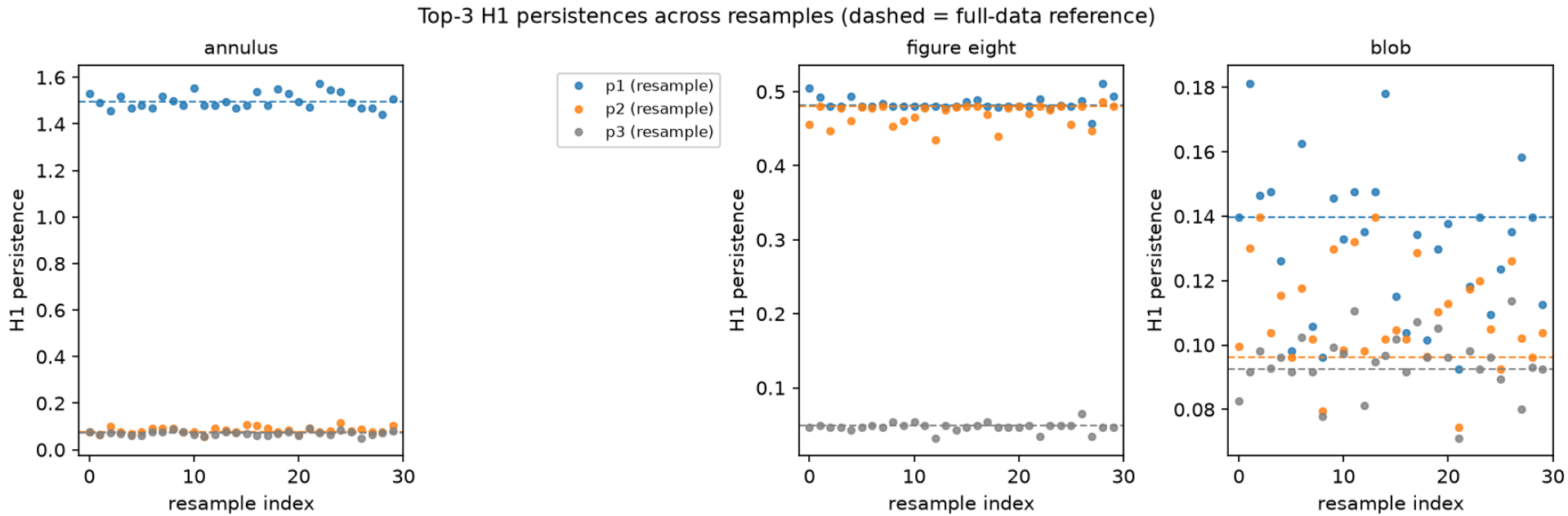
- One 0.57 MB HTML file; 35 saved cells in 4 datasets; shows saved results only, recomputes nothing
- Cold-cache, network-disabled Chrome test: 222/222 checks — zero requests, zero errors, every cell and export matches the source
- Layouts checked from 390 to 1920 px wide; ∞ class drawn as a marked convention
- Not a new algorithm and not the first Mapper explorer




Weather cell (20, 0.35): 2 components, cyclomatic 11, vs the reference cell

# Persistence evidence for a loop holds in every sample — with caveats

- Annulus $r_1 \geq 13.2$ and figure-eight $r_2 \geq 7.35$ in all 60 subsamples; the blob never reaches 3
- Independent redraws: H1 bottleneck median 0.041 vs 0.029 for overlapping subsamples — overlap understates variability
- The relative 1.9% / 4.0% / 34% contrast mostly reflects the sizes of the reference features
- Leave-one-year-out: $r_1$ 4.376–7.754, but 10 of the 36 fold-configurations are effectively two-year omissions




Dominant H1 persistence and gap diagnostic over 30 × 80% subsamples per shape

# What to do with a topological picture

- Ask which representation carries the claim: the diagram, the graph, or the drawing
- Treat Mapper structure as lens-, cover-, **clusterer- and scale-**dependent until shown otherwise for your data
- Use a null that shares the construction (surrogates), not a different representation
- State thresholds as conventions; report distributions over resamples, not one reference
- A graph cycle is not a loop, and a missing cycle is not a missing loop

# Limits and non-claims

- No participant study; E3 is one observer with an AI viewing aid
- One lens, one cover family, two clusterer families; 2-D synthetic shapes and one station
- No cross-implementation check of ripser; no ranking against PCA, t-SNE or UMAP
- Overlapping resamples are descriptive; no p-values or confidence sets; the gap ratio and S are custom, not canonical
- No evidence fixes the order of design and analysis: the sweep was designed knowing the annual period, the seasonality check was post hoc, and the clusterer scales were chosen after the default result
- The weather result is not a climate claim

# Reproducible and checked from outside

- Pinned Python 3.12 stack; results computed on Linux and independently re-executed on Windows 11
- Structural outputs, diagrams and counts identical across platforms; figures pixel-identical; floats within 1e-14
- One command runs every test suite and verification step and fails on any problem
- An evidence verifier re-derives the numbers in the report and slides from the result files
- **Take-away:** trust follows the construction — know which representation, which parameters, and which null